

The Generalized Busemann Intersection Question in Dimension Two

Candidate complete fixed-dimension resolution for arXiv:2208.03882

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Status: candidate substantial partial result, likely valid. Question 1 of the source has an affirmative answer in dimension $n = 2$ for every real $0 < p < 2$. For $p \neq 1$, equality holds exactly for centered ellipses; at the Parseval value $p = 1$, every origin-symmetric star body is an equality case, as the source notes. The higher-dimensional question remains open. Novelty is plausible but not certified.

1 Source question and normalized form

The source asks whether two sharp affine Fourier inequalities hold for all origin-symmetric star bodies, with the direction reversing at $p = n/2$ [1]:

Question 1. *Let $0 < p < n$ be a real number and K be an origin-symmetric star body in $\mathbb{R}^n$. Are the following inequalities true?*

$$\int_{S^{n-1}} \left| (\|\cdot\|_K^{-n+p})^\wedge(\theta) \right|^{n/p} d\theta \leq c_{n,p} (\text{vol}_n(K))^{n/p-1}, \quad \text{if } p < n/2, \quad (6)$$

and

$$\int_{S^{n-1}} \left| (\|\cdot\|_K^{-n+p})^\wedge(\theta) \right|^{n/p} d\theta \geq c_{n,p} (\text{vol}_n(K))^{n/p-1}, \quad \text{if } p > n/2, \quad (7)$$

with equality if and only if K is a centered ellipsoid.

The case $p = n/2$ is omitted above since these inequalities become equalities for every origin-symmetric star body K . This is just an application of the spherical version of Parseval's formula (see [K] p.66):

Let $d\sigma$ be normalized measure on $\mathbb{S}^{n-1}$. The source introduces the normalized spherical Fourier operator I_p , with $I_p 1 = 1$, and shows that for $0 < p < n/2$ the upper inequality is equivalent to

$$\|I_p f\|_{n/p} \leq \|f\|_{n/(n-p)}, \quad f = \rho_K^{-p}. \quad (1)$$

The lower inequality for $n/2 < p < n$ is the reverse norm inequality with the same exponents. Indeed,

$$\int_{\mathbb{S}^{n-1}} f^{n/(n-p)} d\sigma = \frac{\text{vol}_n(K)}{\kappa_n},$$

and the source constant is chosen so that balls give equality. We prove both statements when $n = 2$.

2 Idea of the proof

Proof Intuition

An even function on $\mathbb{S}^1$ is a function on the projective circle, hence has period π . For $0 < p < 1$, the integral kernel of I_p is $|\cos(\theta - \phi)|^{-p}$. A quarter-turn of the output changes this to $|\sin(\theta - \phi)|^{-p}$; doubling angles then identifies the projective circle with the ordinary unit circle and this kernel with its chordal Hardy–Littlewood–Sobolev kernel. Lieb's sharp one-dimensional HLS theorem is exactly the norm estimate (1). Its conformal extremizers are negative powers of positive quadratic forms on the projective circle, which are precisely the radial powers of centered ellipses.

For $1 < p < 2$, put $q = 2 - p < 1$. The spherical Fourier multipliers give $I_q I_p = \text{Id}$ on even functions. Applying the already proved HLS bound for I_q to $I_p f$ reverses the inequality and settles the second half.

3 The projective-circle HLS identity

Write $u_\theta = (\cos \theta, \sin \theta)$ and regard an even function as a π -periodic function of θ . Let

$$d\mu(\theta) = \frac{d\theta}{\pi} \quad (0 \leq \theta < \pi), \quad a_p = \frac{\sqrt{\pi} \Gamma(1 - p/2)}{\Gamma((1 - p)/2)}.$$

The source's integral formula for I_p specializes, for $0 < p < 1$, to

$$(I_p f)(\theta + \pi/2) = a_p \int_0^\pi |\sin(\theta - \phi)|^{-p} f(\phi) d\mu(\phi). \quad (2)$$

The normalization is exact because the beta integral gives

$$\int_0^\pi |\sin t|^{-p} d\mu(t) = \frac{\Gamma((1 - p)/2)}{\sqrt{\pi} \Gamma(1 - p/2)} = a_p^{-1}. \quad (3)$$

We use the following normalized form of the sharp HLS theorem.

Lemma 1 (Sharp HLS on the projective circle). *Let $0 < p < 1$ and $s = 2/(2 - p)$. For $F, G \in L^s([0, \pi), \mu)$,*

$$a_p \left| \int_0^\pi \int_0^\pi \frac{F(\phi)G(\theta)}{|\sin(\theta - \phi)|^p} d\mu(\phi)d\mu(\theta) \right| \leq \|F\|_s \|G\|_s. \quad (4)$$

For nonzero nonnegative F, G , equality holds exactly when they are proportional and, for some $0 \leq r < 1$ and $\beta \in \mathbb{R}$, have the form

$$F(\theta) = A(1 - 2r \cos(2\theta - \beta) + r^2)^{-(2-p)/2}. \quad (5)$$

Proof. Set $x = 2\theta$ and $y = 2\phi$. Then normalized measure becomes normalized measure on the usual circle and

$$2|\sin(\theta - \phi)| = |e^{ix} - e^{iy}|.$$

Thus (4) is the diagonal sharp HLS inequality in dimension one, transported from $\mathbb{R}$ to the circle by stereographic projection. Lieb's sharp theorem gives the constant and all extremizers [2]; evaluating at $F = G = 1$ and using (3) yields the displayed normalization. The conformal extremizers on the circle become exactly (5) after undoing angle doubling. $\square$

Corollary 2. *For $0 < p < 1$ and every even $f \in L^{2/(2-p)}(\mathbb{S}^1)$,*

$$\|I_p f\|_{2/p} \leq \|f\|_{2/(2-p)}. \quad (6)$$

For nonzero nonnegative f , equality holds exactly for the functions in (5).

Proof. The dual exponent of $2/(2 - p)$ is $2/p$. Apply Lemma 1, use duality, and then use (2); the quarter-turn preserves every L^r norm. The equality statement is inherited from the bilinear theorem. $\square$

4 Resolution in dimension two

Theorem 3. *Question 1 of the source is true when $n = 2$.*

1. *If $0 < p < 1$, inequality (6) holds for every origin-symmetric star body $K \subset \mathbb{R}^2$, with equality exactly for centered ellipses.*
2. *At $p = 1$, the Parseval identity gives equality for every origin-symmetric star body.*
3. *If $1 < p < 2$, inequality (7) holds on its natural Fourier domain: if $I_p(\rho_K^{2-p}) \in L^{2/p}$, the displayed integral is finite and the sharp lower bound holds; otherwise its extended value is $+\infty$. Among finite cases, equality holds exactly for centered ellipses.*

Proof. First suppose $0 < p < 1$ and set $f = \rho_K^{2-p}$. Then f is positive, continuous, and even, while

$$\int_{\mathbb{S}^1} f^{2/(2-p)} d\sigma = \int_{\mathbb{S}^1} \rho_K^2 d\sigma = \frac{\text{area}(K)}{\pi}.$$

Corollary 2 is therefore precisely the normalized form (1) of source inequality (6).

It remains to translate equality. The denominator in (5) is a positive quadratic form in u_θ :

$$1 - 2r \cos(2\theta - \beta) + r^2 = u_\theta^T Q_{r,\beta} u_\theta,$$

where

$$Q_{r,\beta} = (1 + r^2)I - 2r \begin{pmatrix} \cos \beta & \sin \beta \\ \sin \beta & -\cos \beta \end{pmatrix}.$$

Its eigenvalues are $(1 - r)^2$ and $(1 + r)^2$, so it is positive definite. Consequently equality means

$$\rho_K(\theta) = C(u_\theta^T Q_{r,\beta} u_\theta)^{-1/2},$$

which is exactly the radial function of a centered ellipse. Conversely, every centered ellipse has this form and gives equality.

At $p = 1$, the source's spherical Parseval identity applies and all bodies are equality cases.

Finally let $1 < p < 2$, set $q = 2 - p$, and initially take smooth even f . On spherical harmonics of degree $2j$, the normalized Fourier multiplier of I_p is

$$(-1)^j \frac{\Gamma(1 - p/2)\Gamma(j + p/2)}{\Gamma(p/2)\Gamma(j + 1 - p/2)}.$$

The multiplier for I_q is its reciprocal, hence $I_q I_p f = f$. Applying (6) with parameter q to $h = I_p f$ gives

$$\|f\|_{2/(2-p)} = \|I_q h\|_{2/(2-p)} \leq \|h\|_{2/p} = \|I_p f\|_{2/p},$$

which is the normalized form of source inequality (7). The same argument holds distributionally whenever $I_p f \in L^{2/p}$; standard approximation gives the stated natural-domain formulation.

If equality holds, the HLS classification for q says $h = C(u_\theta^T Q u_\theta)^{-p/2}$. The HLS Euler–Lagrange equality, or direct conformal covariance, then gives $f = I_q h = C'(u_\theta^T Q u_\theta)^{-(2-p)/2}$, so K is a centered ellipse. The converse also follows from the source's linear invariance and its ellipsoid equality lemma. $\square$

5 Scope and higher-dimensional obstruction

The proof is special to dimension two. On the projective circle, the orthogonality set of a point is another point, so a quarter-turn converts the generalized cosine kernel into the diagonal HLS singularity. In $\mathbb{RP}^{n-1}$ for $n > 2$, the orthogonality set is a hypersurface, not the diagonal, and this reduction disappears.

The later stochastic-geometry theorem of Adamczak, Paouris, Pivovarov, and Simanjuntak proves the $0 < p < 1$ portion corresponding to $n/p \in \mathbb{N}$ after translating their negative L_p -intersection-body parameter [3]. It does not cover continuous p , while the circle argument does. Attempts to interpolate those discrete parameters, to use the ordinary higher-dimensional HLS theorem, and to force an axisymmetric counterexample did not resolve $n \geq 3$; these attempts are recorded with the packet.

References

- [1] V. Yaskin, *On a generalization of Busemann's intersection inequality*, J. Funct. Anal. **287** (2024), 110561; [arXiv:2208.03882](#).
- [2] E. H. Lieb, *Sharp constants in the Hardy–Littlewood–Sobolev and related inequalities*, Ann. of Math. (2) **118** (1983), 349–374.
- [3] R. Adamczak, G. Paouris, P. Pivovarov, and P. Simanjuntak, *From intersection bodies to dual centroid bodies: a stochastic approach to isoperimetry*, J. Eur. Math. Soc. (2024); [arXiv:2211.16263](#).